

HAT-P-12b

R-1012

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_160511 · created 2026-07-30T03:04:47+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457519.7939 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.114 +/- 0.015
Transit depth (Rp/Rs)^2	1.29 +/- 0.35 [%]
Orbital Inclination (inc)	87.82 +/- 0.98
Airmass coefficient 1 (a1)	0.981 +/- 0.014
Airmass coefficient 2 (a2)	0.0155 +/- 0.0088
Scatter in the residuals of the lightcurve fit is	1.39 %
Best Comparison Star	#3 - [124, 347]
Optimal Aperture	3.46
Optimal Annulus	8.22
Transit Duration (day)	0.0869 +/- 0.009

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.114 $\pm$ 0.015 vs 0.1406 (deviation 1.03 σ)
Duration fit vs published	0.0869 d vs 0.0975 d (deviation 0.68 σ)
Mid-time O–C	-4.7 min
Depth SNR	4.4
Residual scatter	1.39 %

Light curve

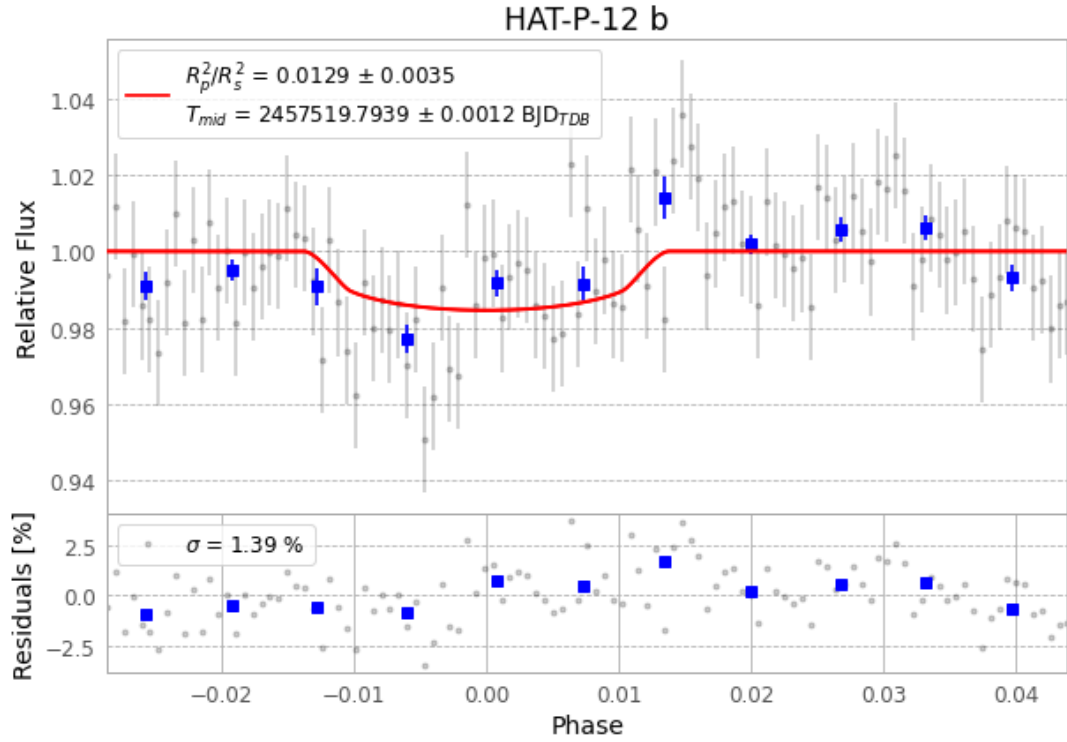


Figure 1 — FinalLightCurve_HAT-P-12 b_2016-05-10.png (SHA-256 6c2c9e4b4e0da47b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [271, 306], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e4a488df7e3fe94d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

113 FITS frames (74 MB), HATP-12160511044512.FITS → HATP-12160511102112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-160511.log (54271a42c8f0...), FinalLightCurve_HAT-P-12 b_2016-05-10.png (6c2c9e4b4e0d...), FinalLightCurve_HAT-P-12 b_2016-05-10.csv (1ad5aa9fec8d...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 03:04Z.